



Global Forum on **PRODUCTIVITY**

Boosting growth and productivity:
the role of human capital

Japan's AI Policy

- Striking Balance between
Promotion of Innovation and
Securing Guardrails -

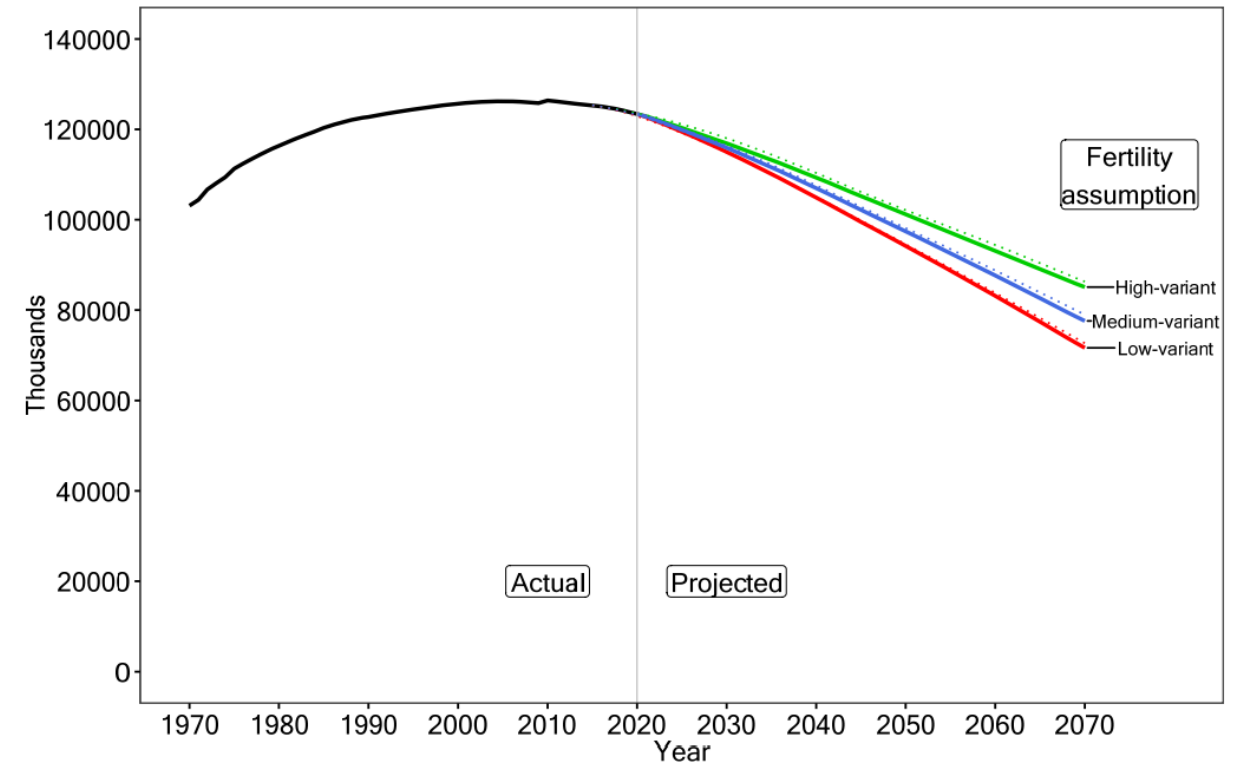
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Big picture

- Population **Declining on the way**

Japanese Population Projections Figure 1 Actual and projected Japanese population: Medium-, high-, and low-fertility (medium-mortality) projections



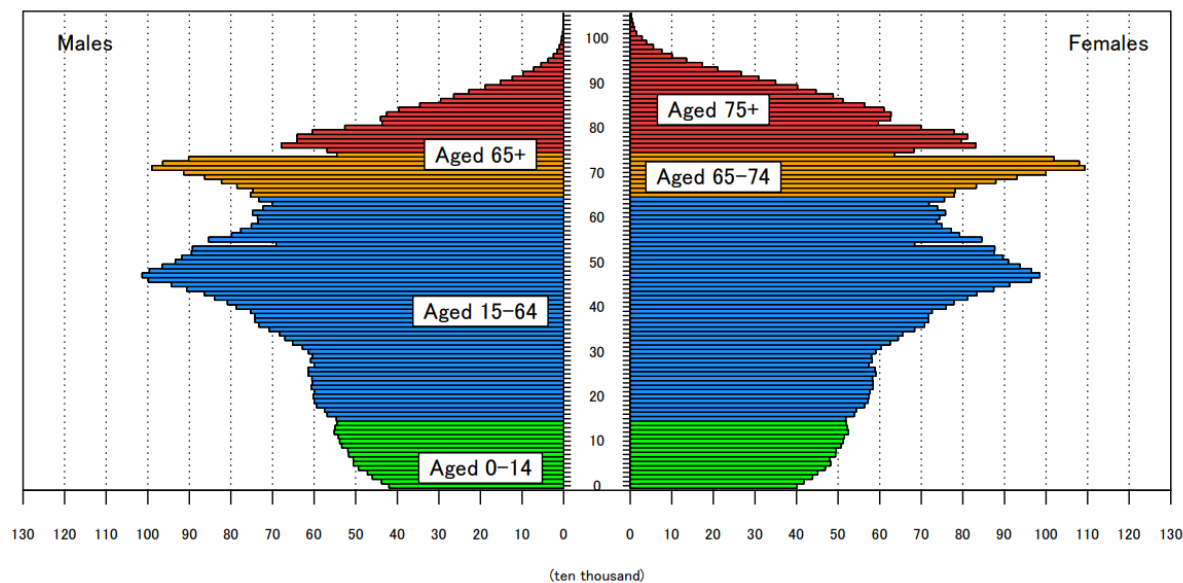
The latest projections are shown in solid lines and previous projections are shown in broken lines.

Source: Statistics Bureau of Japan

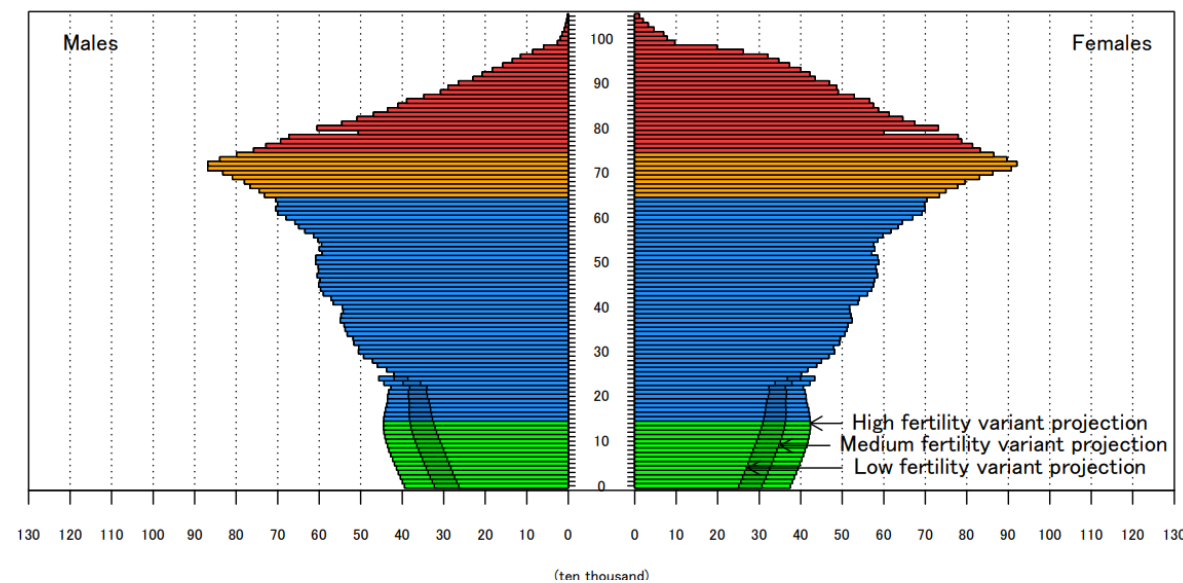
Big picture

- **Rapid decrease of Workforce in couple of decades**

(1) 2020 (Japanese Population)



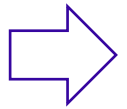
(2) 2045 (Japanese Population)



Source: Statistics Bureau of Japan

Big picture

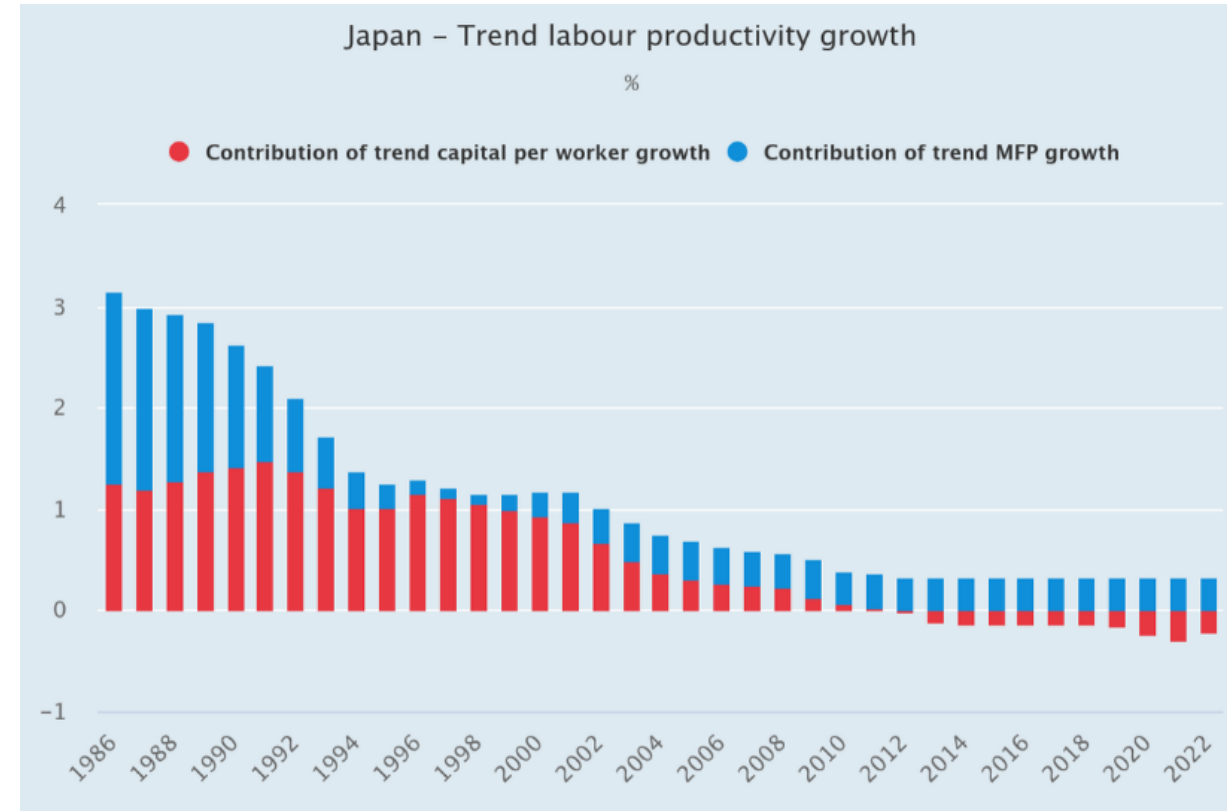
● Decreasing and Burdening Labor Productivity



Urgent necessity to leverage AI to stimulate
Japanese Economics / Labor Market

Ref. OECD Recommendation

- Boost productivity growth by strengthening smaller firms
- Promote a competitive business environment by creating a favourable investment climate and spurring business dynamism
- Enhance labour mobility between businesses, universities, and public research institutions to spur R&D.



Source: OECD

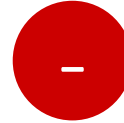
Big picture

Opportunity



- Urgent necessity to improve labor productivity
- Data housed within each industrial sector/ each enterprise

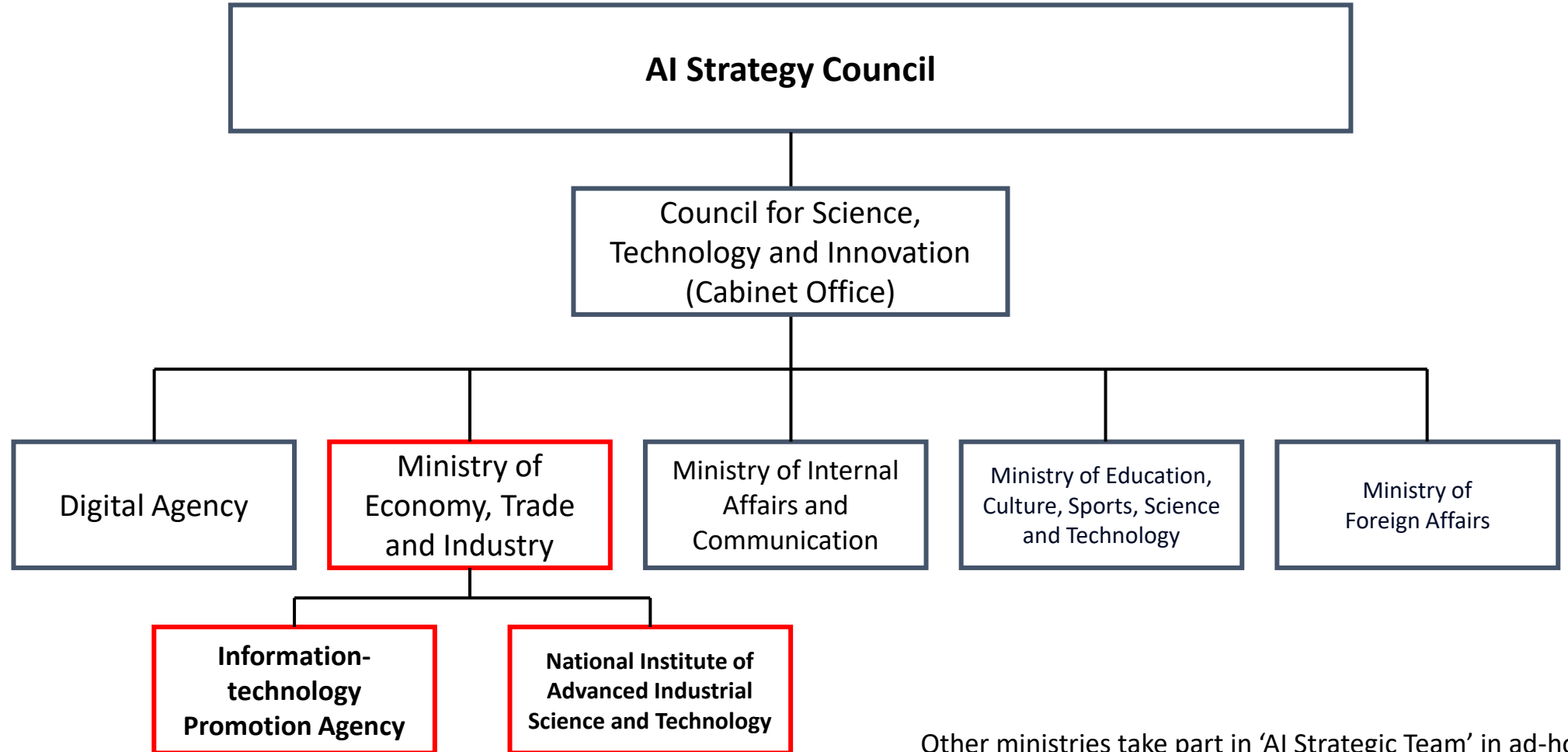
Challenge



- Lack in Digital Talents both in Service Users and Providers
- Lack in agile/ speedy adoption before risks caused by novel technologies

Source: Statistics Bureau of Japan

Policy Approach as a whole



Other ministries take part in 'AI Strategic Team' in ad-hoc manner

Policy Approach as a whole

Basic Concept

Play a leading role in international rule-making/ Striking balance between AI Promotion and Response to AI Risks/ Agile and Flexible response in a multi-stakeholder approach

Response to AI risks

- Contribution to International rule-making
 - **Integration of Existing three AI Guideline to 'AI Guidelines for Business'** (with MIC)
 - **G7 Hiroshima AI Process: International rule making** (with MIC)
 - **AI Safety Institute**
- R&D in counter measures against mis-/dis information

Promotion of AI use

- Promotion of AI use in healthcare, education and infrastructure
- Improvement of AI skills and literacy
 - **Revision of 'Digital Skill Standards'**
 - Investment in human resources in the communities

Investment in AI R&D

- Investment in computational resources
 - **Development of computing resources for AI**
 - **Securing private entities' access to computing resources at AIST** (AI Bridging Cloud Infrastructure)
 - **Support for AI development acceleration**
- Securing access to high-quality data
- Strengthening R&D capabilities in transparent and trustworthy foundation model

AI as Society-ready Technology – Adopting AI to the Society

Generative AI Accelerator Challenge (GENIAC)

Support the private sector and others for their **originality and ingenuity in the development of fundamental models** to strengthen the development capability of generative AI

Pillar 1: Provision of Computing Resources: Support procurement and subsidize usage fees for computing resources, which are a particular challenge in developing the basic model

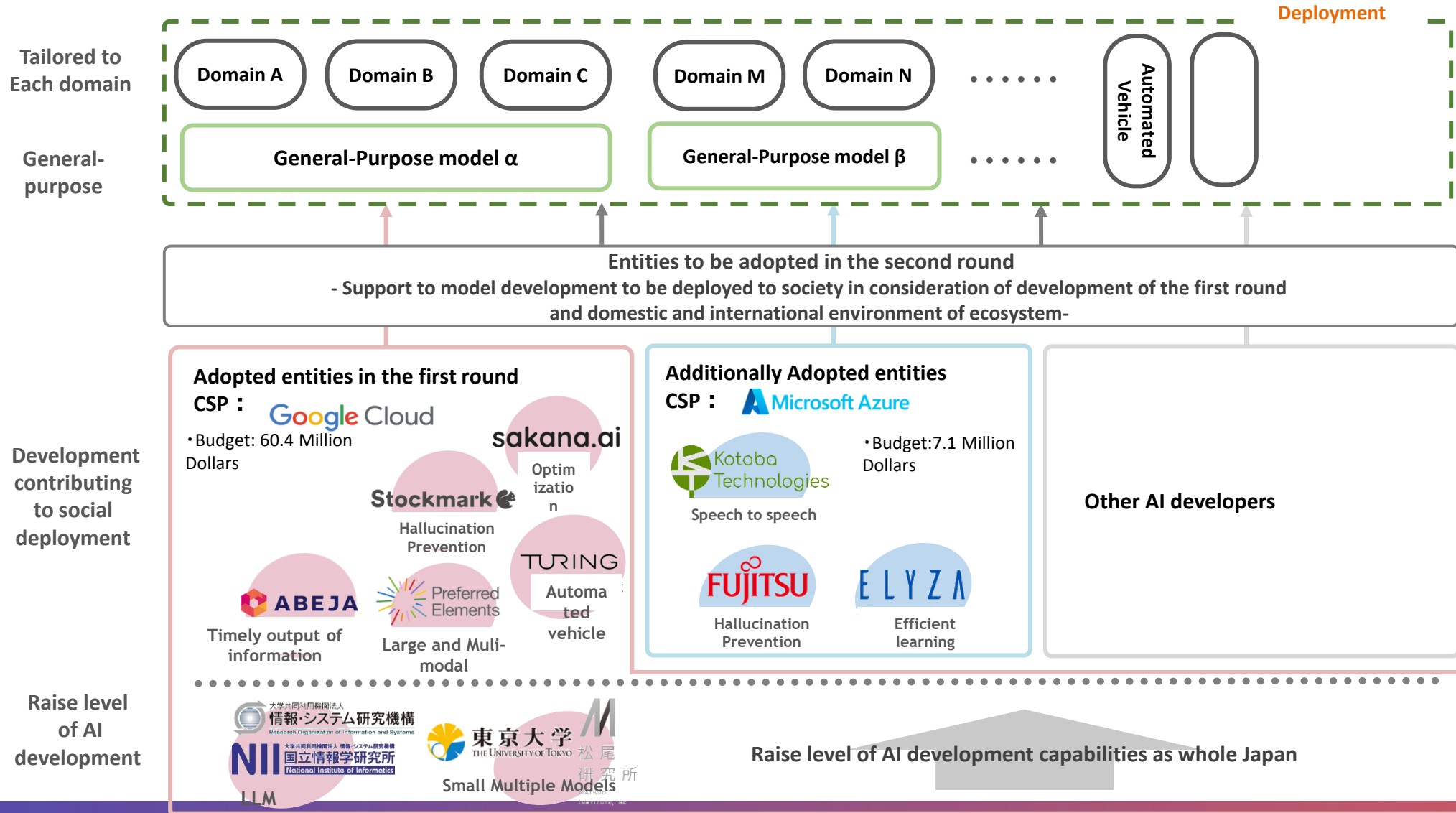
Pillar 2: promotion of exchange among developers: Establish a community that aims to create an environment where developers can share their knowledge, expand their network, and connect their development models to the large number of users

Pillar3: promotion of collaboration of AI developers with AI users and data holders:

- ✓ Utilization of data held by companies and other entities is important to improve the performance and promote the use of generative AI. METI will promote collaboration between data providers and AI developers to resolve issues in data maintenance and regulatory aspects.
- ✓ Specifically, METI will investigate how to construct useful datasets for AI developers and advanced case studies to solve problems in the development and utilization of AI.

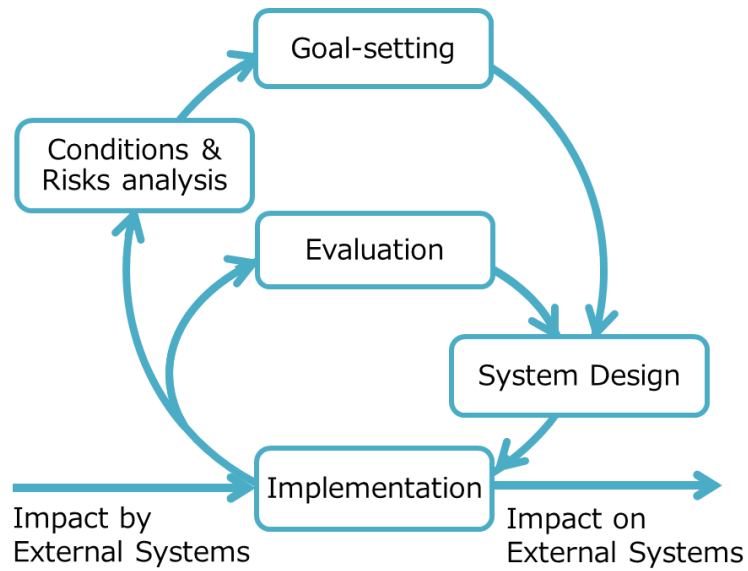


Face-to-face meeting event on February 9th



Society as AI-Ready Instrument – Adjusting Society to the AI: Concept of ‘Agile Governance’

- Necessity of governance measure enabling its improvement in an agile and flexible manner to promote innovation in disruptive and emerging technologies
- Japan’s emphasis on Agile Governance since the publication of ‘Governance Innovation –Resigning Law and Architecture for Society 5.0’ (July, 2020)
- OECD’s ‘Recommendation of the Council for Agile Regulatory Governance to Harness Innovation’ (October, 2021)



Summary of ‘Governance Innovation –Resigning Law and Architecture for Society 5.0’

- Governance of **cyber-physical systems (CPSs)** is necessary for Society5.0 (Ch.1)
- CPSs are complex, change rapidly, and pose **difficulties with respect to controlling their risks** (Ch.2)
- **Goals of governance will also constantly change** in accordance with the changes that these systems undergo (Ch.3)
- We should implement ‘**Agile Governance**’ which is to **continuously and rapidly** run cycles as shown below with **multi-stakeholders** in various governance systems. (Ch.4)

Society as AI-Ready Instrument – Adjusting Society to the AI: AI Guidelines for Business

- Japan's position: **Striking balance between AI Promotion and Response to AI Risks**
-> **Take 'soft law' approach** by guidelines to encourage stakeholders to take voluntary effort (e.g. build internal AI governance systems)
- Basic concept of AI Guidelines for Business
 - 1) **harmonization with international discussions**: reference to existing principles and frameworks of other countries/region and international organizations
 - 2) **multi-stakeholder approach**: balanced discussions among industries, academia and civil society
 - 3) **readability for all AI stakeholders**: not only executives but also persons in practical business of AI developers, providers and users as readers expected

1) Harmonization with international discussions

Refer to the following international and/or internationally recognized documents to extract common elements

e.g. Ethics guidelines for trustworthy AI (2019, EU), Recommendation of the Council on Artificial Intelligence (2019, OECD), Recommendation on the Ethics of Artificial Intelligence (2021, UNESCO), Blueprint for an AI Bill of Right (2022, US), AI Risk Management Framework (2023, US), Draft of AI Act (2023, EU) and multiple deliverables from Hiroshima AI Process

2) Multi-stakeholder approach

Invite more than 100 stakeholders from industries*, academia and civil society to the discussion

* Call on a wide variety of enterprises, considering about their sizes and their relations to AI. Enterprises include not only major Japanese companies (e.g. AI developers and banks), but also non-Japanese companies and Japanese startups.

3) Readability for all AI stakeholders

Concise main body, detailed appendix to encourage all the AI-involved stakeholders can read (See next page)

Society as AI-Ready Instrument – Adjusting Society to the AI: Liability allocation in the era of AI

- Social problems: **Traffic accidents rooted in mistakes by human drivers** (i.e. pedal error by the elderly)
- **Serious lack of human resources** in public transportation of regional communities

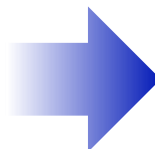
-> More expectation on **early social deployment of safe and secure automated vehicles systems**

To advance further its social deployment, leverage societal rule-makings in addition to existing policy effort including R&D

Current Status / Challenges

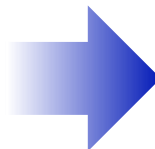
✓ **Improve Predictability of Administrative, Civil and Criminal Liabilities**

- Unclear allocation of liabilities among entities in the value chain (i.e. developers and service operators) in case of accidents of Unmanned Automated Vehicles



✓ **Mechanism to investigate Causes of Accidents**

- Independent Organization with technical expertise (ref. Investigation Committee on Automated Vehicles Accidents within ITARDA)



Targeted Goal

- ✓ **Concrete and Quantitative Standards and Guidelines** to make clear traffic rules of Automated Vehicles
- ✓ **Build up Information-sharing mechanism** to update Quantitative Standards and Guidelines in an agile manner
- ✓ **Design and Operation of Legal Systems** for appropriate consideration about the fact that each Automated Vehicle is in line with Standards and Guidelines in judging liabilities

- ✓ **Establish an Accident Investigation Organization with Jurisdiction and Independency** (specific to Automated Vehicle Accidents) similar to Japan Transport Safety Board